

ROUTE 3 (PATH 3) — COMPLETE ORGANIZER ANSWER KEY & OPERATIONS

GUIDE

CONFIDENTIAL • OFFICIAL ORGANIZER MASTER MANUAL

ROUTE: MBA → ADMIN → ECE → LIBRARY → FOOD COURT → AUDI

 **ORGANIZER INSTRUCTION:** This official manual corresponds to PATH3_FINAL_ODK.x1sx. All typed answers strictly enforce **MANDATORY UPPERCASE**.

1. MASTER FLOW & VOLUNTEER PASS CODES

Stage	Location	Challenge	Variant_Count	Expected_Answer_or_Code	Time_Limit	Unlocks	Notes
Stage 1: MBA Hidden Object	MBA Zone	 Hidden Object — Visvesvaraya	1 (Universal)	MBA-VISV-PASS	3–5 mins	Location Clue (Heart of Governance)	Locate physical object -> Volunteer verifies and gives pass code
Stage 2: Admin Location Clue	Discovery	"Heart of Governance & Central Command"	1 (Universal)	ADMIN	2–3 mins	NEXT BLOCK: ADMIN & Start Code	Decipher campus riddle -> Unlocks ADMIN
Stage 3: Admin Start	Admin Block	Arrival at Admin Station	1 (Universal)	ADMIN-START	Immediate	Admin Phase 1 (Elemental Encryption)	Volunteer assigns Variant A–G
Stage 4: Admin Phase 1	Admin Block	Elemental Encryption (Periodic Table Symbols)	7 Variants (A–G)	CLASS / POWER / SPARK / BRAIN / BOUND / CORNER / OFFICE	3–4 mins	Admin Phase 2 (Count to Unlock)	Decode atomic numbers to elemental letters
Stage 5: Admin Phase 2	Admin Block	Count to Unlock (Hidden Object Counting)	7 Variants (A–G)	14 / 12 / 15 / 11 / 13 / 16 / 10	3–4 mins	Admin Phase 3 (Piece-by-Piece)	Inspect schematic image and count target icons
Stage 6: Admin Phase 3	Admin Block	Piece-by-Piece (Fragmented Assembly Recognition)	7 Variants (A–G)	COMPASS / TURBINE / TELESCOPE / GENERATOR / PROPELLER / MICROSCOPE / CALIPER	4–5 mins	Admin Phase 4 (Paper Ball Relay)	Reconstruct jigsaw pieces and enter object name
Stage 7: Admin Phase 4	Admin Block Courtyard	Paper Ball Relay (4 boards, 1 paper ball, target bucket)	1 (Universal)	ADMIN-RELAY-PASS	4–6 mins	NEXT BLOCK: ECE & Start Code	Physical teamwork challenge verified by volunteer
Stage 8: ECE Start & Riddle	ECE Block Entrance	Electronics Riddle ("I deal in waves, both low and high...")	1 (Universal)	ANALOG LAB	2–3 mins	ECE Phase 2 (School-Level Quiz)	Solves electronics riddle
Stage 9: ECE Phase 2	ECE Block	School-Level Science & Circuits Quiz (Class 11/12)	7 Sets × 5 Qs = 35 Qs	See Quiz Bank Sheet	3–4 mins	ECE Phase 3 (QR Hunt with Decoys)	All 5 questions mandatory, solved sequentially
Stage 10: ECE Phase 3	ECE Building Floors	QR Hunt with Decoys (Authentic + Decoy QR Codes)	7 Route Variants	ECE-QR-A71 / ECE-QR-B82 / ECE-QR-C93 / ECE-QR-D14 / ECE-QR-E25 / ECE-QR-F36 / ECE-QR-G47	5–8 mins	ECE Phase 4 (Building-Wide Hunt)	Navigate to waypoint, avoid decoys, scan authentic QR
Stage 11: ECE Phase 4	ECE Building Labs & Corridors	Building-Wide Physical Navigation & Observation	7 Route Variants	INTEGRATION / MODULATION / FREQUENCY / OSCILLATOR / WAVELENGTH / AMPLITUDE / RESONANCE	6–10 mins	ECE Volunteer Verification	Movement + observation across Ground, 1st & 2nd floors
Stage 12: ECE Verification	ECE Station Desk	Volunteer Verification of Building Hunt	1 (Universal)	ECE-PASS	1–2 mins	NEXT BLOCK: LIBRARY & Caesar Stages	Volunteer inspects solution and gives ECE-PASS
Stage 13: Library Caesar 1-5	Central Library	5 Sequential Caesar Cipher Decryptions	5 Stages (Universal)	CAMPUS (+3) -> CATALOG (+4) -> JOURNAL (+2) -> KNOWLEDGE (+5) -> DIGITALARCHIVE (+1)	5–7 mins	Library Volunteer Verification	All 5 stages mandatory; no skipping
Stage 14: Library Verification	Central Library Desk	Volunteer Verification of Caesar Decipherment	1 (Universal)	LIB-CAESAR-PASS	1–2 mins	NEXT LOCATION: FOOD COURT & Cup Stack	Volunteer verifies all 5 ciphers and issues pass code
Stage 15: Food Court Cup Stack	Food Court Plaza	Colour Cup / Cup Stack Coordination Challenge	1 (Universal)	FC-STACK-PASS	4–6 mins	NEXT BLOCK: AUDI	Physical stacking challenge; volunteer validates and gives code
Stage 16: Audi 1 (Riddle)	Main Auditorium	Auditorium Riddle ("A hall of echoes, where voices rise...")	1 (Universal)	AUDITORIUM	2–3 mins	Audi 2 (Stage Riddle)	Universal final riddle
Stage 17: Audi 2 (Stage)	Main Auditorium Stage	Stage Riddle ("Raised above the wooden floor...")	1 (Universal)	STAGE	2–3 mins	Final Physical Challenge	Universal stage riddle
Stage 18: Final Physical Trial	Grand Stage	Final Physical Coordination & Agility Challenge	1 (Universal)	Final Answer 1: STAGE / Final Answer 2: FINAL-PATH3	5–8 mins	 PATH 3 COMPLETE	Judges verify trial and issue final 2 pass answers

2. ADMIN STAGE (ELEMENTAL, OBJECT COUNT & PUZZLE PIECE)

Variant	P1_Elemental_Formula	P1_Elemental_Answer	P1_Explanation	P2_Count_Item	P2_Count_Answer	P2_Image_File	P3_Piece_Object	P3_Piece_Hint	P3_Image_File	P4_Paper_Ball_Relay_Code
A	[15] P + [53] I + [11] Na + [9] F + [57] La	PINAFLA	P (Phosphorus) + I (Iodine) + Na (Sodium) + F (Fluorine) + La (Lanthanum) = PINAFLA	Golden Key Icons	14	admin_count_A.png	COMPASS	Navigational instrument with magnetized needle pointing north	admin_piece_A.png	ADMIN-RELAY-PASS
B	[6] C + [18] Ar + [9] F + [53] I + [19] K	CARFIK	C (Carbon) + Ar (Argon) + F (Fluorine) + I (Iodine) + K (Potassium) = CARFIK	Compass Rose Icons	12	admin_count_B.png	TURBINE	Rotary mechanical device extracting energy from fluid flow	admin_piece_B.png	ADMIN-RELAY-PASS
C	[7] N + [8] O + [15] P + [53] I + [7] N	NOPIN	N (Nitrogen) + O (Oxygen) + P (Phosphorus) + I (Iodine) + N (Nitrogen) = NOPIN	Gear Cogwheel Icons	15	admin_count_C.png	TELESCOPE	Optical instrument using lenses and mirrors to observe distant objects	admin_piece_C.png	ADMIN-RELAY-PASS
D	[16] S + [9] F + [15] P + [53] I + [7] N	SFPIN	S (Sulfur) + F (Fluorine) + P (Phosphorus) + I (Iodine) + N (Nitrogen) = SFPIN	Magnifying Glass Icons	11	admin_count_D.png	GENERATOR	Electromagnetic machine converting mechanical energy into electricity	admin_piece_D.png	ADMIN-RELAY-PASS
E	[92] U + [7] N + [53] I + [8] O + [7] N	UNION	U (Uranium) + N (Nitrogen) + I (Iodine) + O (Oxygen) + N (Nitrogen) = UNION	Padlock Security Icons	13	admin_count_E.png	PROPELLER	Rotating aerodynamic device with blades providing forward propulsion	admin_piece_E.png	ADMIN-RELAY-PASS
F	[27] Co + [86] Rn + [68] Er	CORNER	Co (Cobalt) + Rn (Radon) + Er (Erbium) = CORNER	Star Insignia Tokens	16	admin_count_F.png	MICROSCOPE	Laboratory optical instrument magnifying microscopic specimens	admin_piece_F.png	ADMIN-RELAY-PASS

Variant	P1_Elemental_Formula	P1_Elemental_Answer	P1_Explanation	P2_Count_Item	P2_Count_Answer	P2_Image_File	P3_Piece_Object	P3_Piece_Hint	P3_Image_File	P4_Paper_Ball_Relay_Code
G	[8] O + [9] F + [9] F + [53] I + [58] Ce	OFFICE	O (Oxygen) + F (Fluorine) + F (Fluorine) + I (Iodine) + Ce (Cerium) = OFFICE	Anchor Emblem Tokens	10	admin_count_G.png	CALIPER	Precision dual- jaw measuring instrument with vernier scale	admin_piece_G.png	ADMIN-RELAY-PASS

3. ECE LOGIC & CIRCUIT BREAKER QUESTION BANK

Set	Q_Number	Question_Text	Expected_Answer	Academic_Level	Case_Rule
A	1	What is the SI unit of electrical resistance?	OHM	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
A	2	What device is connected in series in a circuit to measure electric current?	AMMETER	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
A	3	Which scientist formulated the law of electromagnetic induction stating induced EMF equals rate of change of flux?	FARADAY	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
A	4	In semiconductors, what is the majority charge carrier in n-type semiconductor material?	ELECTRON	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
A	5	What optical phenomenon causes the splitting of white light into constituent colors through a glass prism?	DISPERSION	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
B	1	What is the SI unit of electric capacitance?	FARAD	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
B	2	What instrument is connected in parallel across a component to measure potential difference?	VOLTMETER	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
B	3	What semiconductor component allows electric current to flow primarily in only one direction?	DIODE	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
B	4	What optical lens curves inward and causes parallel light rays to spread apart?	DIVERGING	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
B	5	In physics, which subatomic particle carries a single negative elementary electric charge?	ELECTRON	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
C	1	What is the SI unit of magnetic flux density (magnetic field)?	TESLA	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
C	2	What fundamental circuit law states that potential difference V equals current I multiplied by resistance R?	OHMS LAW	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
C	3	What intrinsic property of a conductor opposes the flow of electric current?	RESISTANCE	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
C	4	What wave phenomenon describes the bouncing back of a light ray from a reflective surface?	REFLECTION	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
C	5	What tetravalent metalloid element with atomic number 14 is the primary semiconductor base material for microchips?	SILICON	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
D	1	What is the SI unit of wave frequency, equal to one cycle per second?	HERTZ	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
D	2	What passive component stores electrical energy directly in an electrostatic field between two plates?	CAPACITOR	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
D	3	In semiconductors, what are the majority charge carriers in p-type semiconductor material?	HOLES	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
D	4	What optical phenomenon enables light signals to travel without escaping through fiber optic cables?	TOTAL INTERNAL REFLECTION	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
D	5	What electromagnetic machine converts rotational mechanical power into electrical energy?	GENERATOR	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
E	1	What is the SI unit of electrical power (equal to one Joule per second)?	WATT	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
E	2	What component opposes sudden changes in current by storing energy in a magnetic field coil?	INDUCTOR	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
E	3	What is the reciprocal of electrical resistance called (measured in Siemens)?	CONDUCTANCE	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
E	4	What curved mirror curves outward and always forms a virtual, diminished, upright image?	CONVEX	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
E	5	What is the intentional introduction of impurities into an intrinsic semiconductor called?	DOPING	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
F	1	What is the SI unit of electric charge (symbol C)?	COULOMB	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
F	2	What three-terminal semiconductor component is used to amplify or switch electronic signals?	TRANSISTOR	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
F	3	Which circuit law states that the algebraic sum of currents entering any circuit node is zero?	KIRCHHOFF	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
F	4	What wave phenomenon describes the spreading or bending of waves passing through narrow apertures?	DIFFRACTION	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
F	5	What type of electrical transformer increases voltage from primary to secondary winding?	STEP UP	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
G	1	What is the SI unit of magnetic flux (symbol Wb)?	WEBER	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
G	2	What sensitive electromagnetic device is used to detect the presence and direction of minute electric currents?	GALVANOMETER	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
G	3	What physical quantity is obtained by multiplying voltage V by current I in a DC electrical circuit?	POWER	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
G	4	What optical phenomenon describes the change in direction and speed of light passing across media boundaries?	REFRACTION	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY
G	5	What absolute thermodynamic temperature scale starts at absolute zero (0 K)?	KELVIN	Class 11/12 Physics & Basic Electronics	UPPERCASE ONLY

4. ECE QR & PHYSICAL HUNT DETAILS

Variant	QR_Location	QR_Clue	QR_Code	QR_Image_File	Building_Hunt_Floors	Building_Hunt_Mission	Building_Hunt_Answer	ECE_Final_Volunteer_Code
A	Ground Floor Embedded Systems Board (Near Room 102)	Search the bulletin display opposite Embedded Systems Lab 102.	ECE-QR-A71	ece_qr_A_01.png	Ground Floor -> 1st Floor	Locate the Digital Circuits Lab entrance plaque on Ground Floor. Note the year inscribed, then proceed to 1st Floor Landing to find the highlighted keyword on the wall banner.	INTEGRATION	ECE-PASS
B	First Floor DSP Noticeboard (Opposite Room 204)	Locate the Digital Signal Processing showcase on the 1st Floor East corridor.	ECE-QR-B82	ece_qr_B_01.png	1st Floor -> 2nd Floor	Locate the Communication Lab antenna exhibit on Floor 1. Follow the route arrows to Floor 2 West corridor and identify the bold modulation technique on the technical poster.	MODULATION	ECE-PASS
C	Ground Floor Robotics Arena Display (Near West Exit)	Examine the showcase panel beside the ground floor robotics development hub.	ECE-QR-C93	ece_qr_C_01.png	Ground Floor -> 2nd Floor	Inspect the Robotics showcase on Ground Floor. Count the motor drive channels, then ascend to 2nd Floor East wing to find the keyword displayed above Lab 308.	FREQUENCY	ECE-PASS

Variant	QR_Location	QR_Clue	QR_Code	QR_Image_File	Building_Hunt_Floors	Building_Hunt_Mission	Building_Hunt_Answer	ECE_Final_Volunteer_Code
D	Second Floor Microprocessor Showcase (Room 305)	Ascend to Second Floor West wing and locate the vintage microprocessor display.	ECE-QR-D14	ece_qr_D_01.png	2nd Floor -> 1st Floor	Find the Microprocessor trainer unit on 2nd Floor. Note the crystal clock label, then proceed down to 1st Floor central bulletin to enter the matching circuit block name.	OSCILLATOR	ECE -PASS
E	First Floor VLSI Cleanroom Entrance (Room 210)	Find the wall plaque near the VLSI design and layout testing suite on Floor 1.	ECE-QR-E25	ece_qr_E_01.png	1st Floor -> Ground Floor	Visit the VLSI wafer casing on 1st Floor. Read the laser etching inscription, then proceed to Ground Floor South foyer to read the target optical parameter.	WAVELENGTH	ECE -PASS
F	Ground Floor Antenna & RF Testbench Area (Room 108)	Check the RF laboratory notice column along the central ground floor atrium.	ECE-QR-F36	ece_qr_F_01.png	Ground Floor -> 2nd Floor	Locate the Solar Tracking demonstrator in Ground Floor project bay. Note the servo angle specification, then proceed to 2nd Floor North lobby to get the signal attribute.	AMPLITUDE	ECE -PASS
G	Second Floor Optical & Microwave Lab Entrance (Room 312)	Proceed to Second Floor North corridor near the microwave waveguide enclosure.	ECE-QR-G47	ece_qr_G_01.png	2nd Floor -> 1st Floor	Examine the Microwave waveguide fixture on 2nd Floor. Note the cutoff frequency, then proceed to 1st Floor stairwell landing to identify the circuit tuning principle.	RESONANCE	ECE -PASS

5. LIBRARY CAESAR DECRYPTION & FOOD COURT KEYS

Stage	Field_Name	Shift_Key	Ciphertext	Plaintext_Answer	Clue_Hint	Case_Rule
1	caesar_1_ans	+3	FDPSXV	CAMPUS	Decipher the 6-letter campus keyword using a Caesar shift of +3.	UPPERCASE ONLY
2	caesar_2_ans	+4	GEXEPSK	CATALOG	Decipher the 7-letter library index term using a Caesar shift of +4.	UPPERCASE ONLY
3	caesar_3_ans	+2	LQWTPCN	JOURNAL	Decipher the 7-letter academic publication term using a Caesar shift of +2.	UPPERCASE ONLY
4	caesar_4_ans	+5	PSTBQILJ	KNOWLEDGE	Decipher the 9-letter intellectual concept using a Caesar shift of +5.	UPPERCASE ONLY
5	caesar_5_ans	+1	EJHJUBMBSDIJWF	DIGITALARCHIVE	Decipher the 14-letter library technology system using a Caesar shift of +1.	UPPERCASE ONLY

6. DETAILED CHALLENGE DESCRIPTIONS, MATERIALS & PARALLEL STATION CAPACITIES

Stage 1: MBA Sales Challenge & Quickfire

PARALLEL STATIONS: 4 VERIFICATION DESKS

Event & Mechanics: 25 starting teams receive selling inventory pouches (15 mins campus selling) + solve MBA business quickfire.

Location: MBA Courtyard	Materials: 25 Inventory pouches, cash logs	Staffing: 4 Volunteers
Capacity: 25 Teams (Top 20 qualify)	Pass Code: P3-MBA-PASS	Next Block: Admin Block

Stage 2: Admin Physical Search (Gym Area)

PARALLEL STATIONS: 3 DESKS

Event & Mechanics: 20 teams search Admin outdoor perimeter for physical clue token. Present token to volunteer for verification.

Location: Admin Block Courtyard	Materials: Physical token cards	Staffing: 3 Volunteers
Capacity: 20 Teams (3-5 min each)	Pass Code: P3-ADMIN-PASS	Next Block: ECE Hardware Block

Stage 3: ECE Logic Gate Circuit Breaker

PARALLEL STATIONS: 4 LAB TEST BENCHES

Event & Mechanics: Teams analyze Boolean logic gate matrices (AND/OR/XOR/NAND) across 7 variant sets on laboratory benches.

Location: ECE Hardware Lab	Materials: 4 Circuit test benches, logic sheets	Staffing: 4 Lab Volunteers
Capacity: 20 Teams (Top 15 qualify)	Pass Code: P3-ECE-PASS	Next Block: Central Library

Stage 4: Central Library Book Memory & Catalog Mystery

PARALLEL STATIONS: 4 LIBRARY BAYS

Event & Mechanics: 15 teams observe 16-book visual matrix for 30s and solve missing book mystery.

Location: Library 1st Floor	Materials: Memory matrix boards, stopwatches	Staffing: 4 Volunteers
Capacity: 15 Teams (Top 7 qualify)	Pass Code: P3-LIB-PASS	Next Block: Food Court

Stage 5: Food Court Cryptogram Challenge

PARALLEL STATIONS: 3 VERIFICATION DESKS

Event & Mechanics: 7 teams decode 8 encrypted food & dining words using 4=A...29=Z rule.

Location: Food Court Foyer	Materials: Cryptogram decoder sheets	Staffing: 3 Volunteers
Capacity: 7 Teams (Top 5 qualify)	Pass Code: P3-FC-PASS	Next Block: Main Auditorium

Stage 6: Main Auditorium Grand Finale

PARALLEL STATIONS: 2 STAGE ARENAS

Event & Mechanics: Finalists solve Auditorium riddle (AUDITORIUM), Stage riddle (STAGE), and complete physical coordination trial.

Location: Main Auditorium Stage	Materials: Coordination challenge props	Staffing: 3 Chief Judges
Final Codes: Ans 1: STAGE • Ans 2: FINAL-PATH3	Conclusion: 🏆 PATH 3 COMPLETE	Podium: 1st, 2nd, 3rd Place Award Ceremony